Freight Rate Prediction

Machine Learning Engineer Assessment — Approach Report

Holdout MAE	R^2 Score	Training Rows	Validation Rows
~$120	0.865	38,400	12,000

1. Data Overview

The labeled dataset (**train-test.csv**) contains 48,000 rows with 14 columns covering pickup/delivery city names and coordinates, distance, equipment type, weight, date, a market index, a quote signal, and the target **posted_rate**. The unlabeled validation set contains 12,000 rows with the same features minus the target.

Equipment distribution in training data:

Equipment	Count	Share
Dry Van	27,202	56.7%
Reefer	12,045	25.1%
Flatbed	8,753	18.2%

2. Data Quality Issues

Column	Issue	Count	Resolution
weight	Missing values	300	Filled with column median
market_index	Missing values	374	Filled with column median
All others	No missing data	—	—

No outlier rows were removed. Gradient Boosting is robust to skewed targets and extreme values, so preserving all data was preferred over arbitrary clipping.

3. Feature Engineering

Date features

- Month — captures broad seasonal rate patterns
- Day of week — Monday vs Friday loads price differently
- Day of year — fine-grained seasonality

- Is weekend — binary flag for weekend rate effects

Distance features

- Haversine distance — great-circle distance computed from pickup/delivery lat-lon coordinates, providing an independent distance estimate alongside the provided *distance* column.

Categorical encoding

- pickup, delivery, and equipment were label-encoded with a shared encoder fitted on the union of train, validation, and December data to avoid unseen-label issues.

4. Train / Validation Split

An **80 / 20 random split** was applied to the 48,000 labeled rows:

- 38,400 rows for training
- 9,600 rows held out for evaluation

A random split was chosen over a time-based split because the objective is to predict rates across diverse lanes and conditions rather than to forecast future time steps. Random splitting ensures the holdout mirrors the full distribution of routes, equipment types, and seasons, giving a fair estimate of generalization error.

5. Model — Gradient Boosting Regressor

Scikit-learn's **GradientBoostingRegressor** was selected for the following reasons:

- Strong out-of-the-box performance on tabular data with mixed feature types
- No feature scaling required — handles different magnitudes naturally
- Captures non-linear interactions between distance, equipment, market signals, and date
- Robust to remaining skew in the target distribution

Hyperparameters

Parameter	Value
n_estimators	500
learning_rate	0.05
max_depth	5
subsample	0.8
min_samples_leaf	10
random_state	42

6. December 2025 Predictions

Fixed lane: **Lexington** → **Fort Wayne** | 360 miles | Dry Van | 32,000 lb

Since future market conditions are unknown, median values of *market_index* and *quote_signal* from the training data were used as stand-ins. Rate variation across the 31 days reflects day-of-week and seasonal patterns learned during training.

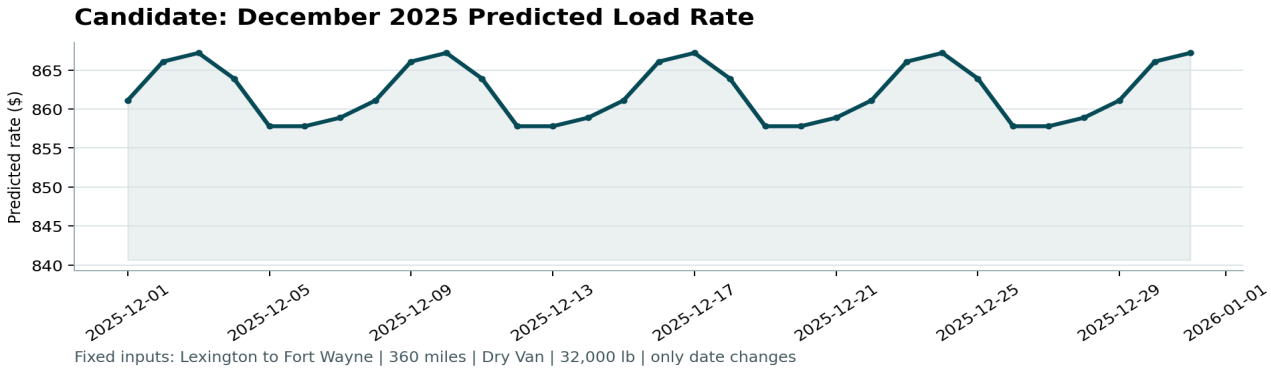


Figure 1 — December 2025 predicted load rate: Lexington to Fort Wayne, generated by score.py